

An Internship Report
Of
Ethical Hacking Virtual Internship (English Language)

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410030690

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, AKSHAY PANDEY, do hereby declare that the internship report titled “Ethical Hacking Virtual Internship (English Language)” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team and EduSkills Academy for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Ethical Hacking. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

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TABLE OF CONTENTS

1. Candidate's Declaration	i
2. Acknowledgement	ii
3. Internship Completion Certificate	-
4. Project Description	1
4.1 Introduction	1
4.2 Organization Profile	2
4.3 Problem Statement	2
4.4 Project Objectives.....	3
4.5 Scope of the Project	3
4.6 Technologies and Tools Used.....	4
4.7 System Architecture.....	4
4.8 Methodology.....	5
4.9 Project: Basic Vulnerability Assessment of a Test Web Application	6
4.10 Expected Outcomes.....	7
4.11 Certificate of Completion and Communication Proof.....	7
5. Bibliography/References.....	8

3. INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
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AICTE
All India Council for Technical Education



NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Akshay Pandey

IILM University, Greater Noida

has successfully completed the 8-weeks

Ethical Hacking Virtual Internship (english Language)

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported by  **EduSkills**
ACADEMY



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education





Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4116a3d90738dec67bb4

Student ID: STU69e258fb715d21776441595



GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of “Ethical Hacking (English Language)”, supported by EduSkills Academy. The internship was carried out from April 2026 to June 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary aim of this internship was to build a strong foundational understanding of ethical hacking and cybersecurity concepts, including the mindset, methodology, and basic tools used by security professionals to identify and analyse vulnerabilities in computer systems and networks in a lawful and controlled manner.

The internship followed a structured, week-wise curriculum combining conceptual learning with hands-on practice, quizzes, and assessments. On successful completion of all the requirements, a Virtual Internship Certificate with an “A” grade was awarded by EduSkills Academy under the patronage of AICTE and the National Internship Portal.

4.2 Organization Profile

EduSkills Academy is an organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. For the Ethical Hacking track, EduSkills designs an industry-aligned curriculum that helps engineering students build a practical understanding of cybersecurity fundamentals and ethical hacking practices.

4.3 Problem Statement

Although many engineering students study the theoretical foundations of networking and computer security during their academic coursework, very few get structured, hands-on exposure to the actual practice of ethical hacking – that is, the lawful identification of vulnerabilities in systems and networks using the same techniques an attacker might use, but with permission and for defensive purposes. The problem addressed by this internship was to build a practical, beginner-level understanding of ethical hacking concepts, tools, and methodology, so that security weaknesses can be identified and reported responsibly.

4.4 Project Objectives

- To understand the fundamental concepts, terminology, and legal/ethical boundaries of ethical hacking.
- To learn the phases of hacking – reconnaissance, scanning, gaining access, maintaining access, and covering tracks – studied from a defensive perspective.
- To understand common network and system vulnerabilities and how they are exploited.
- To get familiar with basic penetration-testing and scanning tools used by security professionals.
- To learn the basics of malware, social engineering, and web-application vulnerabilities.
- To understand the importance of security policies, cryptography basics, and countermeasures.
- To successfully clear the weekly assessments and the Final Assessment Test of the internship.

4.5 Scope of the Project

The scope of this internship spans an introductory, virtual, lab-based overview of ethical hacking – covering networking fundamentals, footprinting and reconnaissance, scanning, system hacking concepts, malware threats, social engineering, and basic web-application and network security. The internship was limited to a simulation/learning-based environment provided by EduSkills Academy and did not involve testing of any live, third-party, or production system.

4.6 Technologies and Tools Used

- Concepts: Networking fundamentals (TCP/IP, ports, protocols), basics of cryptography
- Reconnaissance & Scanning: Basic use of tools such as Nmap for host and port scanning
- Vulnerability Analysis: Identifying common vulnerabilities such as weak configurations, outdated software, and open ports
- Web Security Basics: Introduction to common web-application vulnerabilities (e.g., weak authentication, injection-style issues) at a conceptual level
- Learning Platform: EduSkills Academy structured learning modules and assessments

4.7 System Architecture

The general approach followed during the internship for the Ethical Hacking learning track can be summarised in the following stages:

- Information Gathering Layer: Passive and active reconnaissance to collect information about a target system/network.
- Scanning Layer: Identifying live hosts, open ports, and running services using basic scanning tools.
- Vulnerability Identification Layer: Analysing the gathered information to spot potential weaknesses or misconfigurations.
- Reporting Layer: Documenting findings in a clear, responsible manner, as would be done in a professional penetration-testing report.
- Assessment Layer: Weekly quizzes and a Final Assessment Test tracking progress and final grading.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning and quiz-based assessments, culminating in a Final Assessment Test, as summarised in the table below.

Week	Module	Description
1	Introduction to Ethical Hacking	Overview of information security, ethical hacking concepts, terminology, and the legal/ethical framework governing security testing.
2	Footprinting and Reconnaissance	Techniques for passive and active information gathering about a target system or network.
3	Scanning Networks	Identifying live hosts, open ports, and services using basic network-scanning tools and techniques.
4	System Hacking Concepts	Understanding password attacks, privilege escalation, and system-level vulnerabilities at a conceptual level.
5	Malware Threats	Study of viruses, worms, trojans, and ransomware, along with basic detection and prevention concepts.
6	Social Engineering & Sniffing	Understanding human-factor attacks (phishing, pretexting) and basic network-sniffing concepts.
7	Web Application & Network Security Basics	Introduction to common web-application vulnerabilities and network-security countermeasures.
8	Cryptography & Security Policies	Basics of cryptographic techniques and the role of organisational security policies.
Final	Final Assessment Test	Comprehensive assessment covering all modules; overall grade based on weekly quizzes and this final test.

Note: There was no stipend associated with this internship.

Each week concluded with quiz-based assessments to evaluate conceptual understanding. In the final phase, a Final Assessment Test was conducted, and the overall grade was based on performance across the weekly modules and this final test.

4.9 Project: Basic Vulnerability Assessment of a Test Web Application

As a simple, practical application of the concepts learnt during the internship, a small vulnerability-assessment exercise was carried out on a local, purpose-built test web application (a deliberately vulnerable practice application used only for learning, not a live or third-party system).

Objective: To identify a few common, basic security weaknesses in the test application using simple reconnaissance and scanning steps, and to suggest simple fixes.

Steps followed:

- Set up a local, isolated test environment with a deliberately vulnerable sample web application.
- Performed basic reconnaissance to note the technologies, pages, and forms used by the application.
- Used a basic network scan (Nmap) on the local test server to check open ports and running services.
- Manually checked the login form for a weak/default password and observed how the application responded to an incorrect input.
- Noted down the observations in a short report along with simple recommended fixes (e.g., using strong passwords, closing unused ports, keeping software updated).

Outcome: The exercise helped translate the theoretical modules of the internship into a small hands-on activity, giving a basic first-hand understanding of how a simple vulnerability assessment is carried out and documented in practice. The exercise was kept intentionally simple and was conducted only in a private, local test setup.

4.10 Expected Outcomes

- Ability to explain the fundamental concepts and legal/ethical boundaries of ethical hacking.
- Practical understanding of the basic phases of hacking and common vulnerabilities.
- Familiarity with basic reconnaissance and scanning tools used in security assessments.

- Basic awareness of malware, social engineering, and web-application security concepts.
- Hands-on exposure to conducting a simple vulnerability assessment and documenting findings responsibly.
- Successful clearance of all weekly assessments and the Final Assessment Test, resulting in a Grade “A”.

4.11 Certificate of Completion and Communication Proof

The Internship Certificate of Virtual Internship issued by EduSkills Academy, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Ethical Hacking (English Language) with Grade A, is attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

1. EduSkills Academy, “AICTE – EduSkills Virtual Internship Program – Ethical Hacking (English Language),” Certificate of Virtual Internship, Certificate ID: 4116a3d90738dec67bb4, Student ID: STU69e258fb715d21776441595, 2026.
2. EC-Council, “Certified Ethical Hacker (CEH) Overview,” <https://www.eccouncil.org/>
3. OWASP Foundation, “OWASP Top Ten,” <https://owasp.org/www-project-top-ten/>
4. Nmap Documentation, <https://nmap.org/docs.html>
5. AICTE – National Internship Portal, <https://internship.aicte-india.org/>
6. EduSkills Academy, <https://eduskillsfoundation.org/>